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Polarization Properties of Fused Fiber Couplers and Polarizing Beamsplitters

Katsumi Morishita, *Senior Member, IEEE*, and Katsuyoshi Takashina

Abstract—The polarization properties of fused tapered fiber couplers have been investigated experimentally. It becomes evident from the experimental results that the coupling coefficient difference between x and y polarized lights changes from positive to negative with decreasing the coupler's degree of fusion and the polarization effect is relatively large for weakly fused couplers. Polarizing beamsplitters with the high and the low degrees of fusion are fabricated, and the spectral characteristics are compared with each other.

I. INTRODUCTION

SINGLE-MODE fiber couplers are key components in many sensor and communication applications, including optical power splitting, optical filtering, optical reflecting, wavelength multiplexing/demultiplexing, and optical polarization selective splitting. There is a great interest in the polarization properties of fused single-mode tapered fiber couplers. The polarization behavior of fused fiber couplers has been reported experimentally [1]–[4] and theoretically [5]–[10]. In the measurements and theoretical results, output power exhibits a rapid sinusoidal variation and displays a slow modulation. The reason for the modulation is that the orthogonal polarized modes experience different coupling strengths. Splitting ratio of long fused couplers is influenced particularly by the polarization state of input light [11], [12]. The difference of coupling strengths between x and y polarized lights has been calculated for the elliptic and the dumbbell cross sections [8], [9]. They indicate that the difference changes from positive to negative. However, the difference has not been examined for actual fused taper couplers yet.

This paper provides the first experimental realization of this prediction. The polarization properties of fused tapered fiber couplers are investigated experimentally by changing cross-sectional shapes at the coupler waist. The coupling coefficient difference between x and y polarized lights is studied for many test fused fiber couplers with various cross-sectional shapes. On the basis of these experimental data, polarizing beamsplitters are fabricated and compared with each other to demonstrate the characteristics difference caused by coupler cross sections.

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II. DEPENDENCE OF POLARIZATION PROPERTIES ON CROSS-SECTIONAL SHAPE

In order to investigate the polarization properties of fused single-mode fiber couplers, couplers are fabricated with adjusting the coupler's degree of fusion by change of flame position and temperature. While the flame is scanning back and forth, fused fibers are tapered slowly by pulling the fibers. The width of the heating zone is controlled by moving the flame. The scan length is 5 mm. Butane and oxygen gas flows are less than 3 ml/min and 4–10 ml/min, respectively. The pull speed is 15 $\mu\text{m/s}$. During coupler formation, the optical output power from the coupled port is monitored in real time. We use a laser diode operating at 1.3- μm wavelength to launch the light into the input fiber. The fiber employed in the couplers (SS10A, Mitsubishi Cable Industries, Ltd.) is a conventional single-mode fiber, i.e., a standard nonpolarization-preserving fiber, with mode field diameter of 9.5 μm at 1.3- μm wavelength and fiber diameter of 125 μm . The spectral characteristics of manufactured couplers are measured using a tungsten halogen lamp for monochromator illumination and a germanium photodiode to detect the chopped optical power. The polarized light is launched into the input fiber through a polarizer with extinction ratio of 50 dB. After being characterized, couplers are cleaved at the taper waist and the cross-sectional shape is observed.

Fig. 1 shows the polarization-maintaining characteristics of the employed single-mode fiber. The polarized light through the polarizer is launched into the fiber with a 35 cm length, and the output power through an analyzer is measured rotating it. We can see that the input polarization is preserved to the extent of extinction ratio of 18 dB. Extinction ratio is almost the same for any input polarization angle. The fiber from the input end to the coupling region is cut in a length of about 15 cm to assure that the input polarization is maintained to extinction ratio of more than 18 dB in transmission through the coupler pigtail.

Fig. 2 shows the spectral power transfer difference between x and y polarized lights. The coupled power P_2 schematically indicated varies with wavelength λ . Both polarized lights have the similar pattern of the spectral power transfer. The wavelength difference between their patterns is indicated by Δt . In case of $\Delta t > 0$, the coupling coefficient for the x polarized mode is larger than for the y polarized mode. The aspect ratio l/d is defined

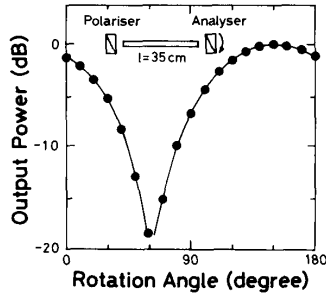


Fig. 1. Polarization-maintaining characteristics of the employed single-mode fiber.

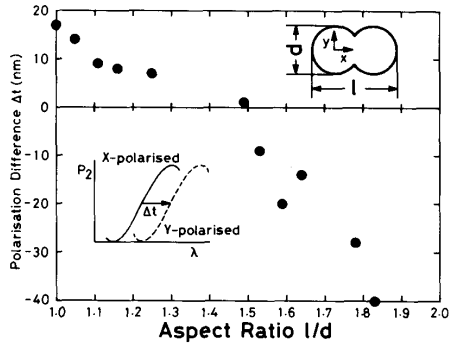


Fig. 2. Spectral power transfer difference between x and y polarized lights. The height and width at the coupler waist are d and l . P_2 is the power transferred from the input fiber to the coupled fiber, and the change of P_2 with wavelength λ is indicated schematically.

by the width l and the height d of the cross section at the coupler waist. The cross sections with the aspect ratios of 1.0, less than about 1.5, and more than about 1.5 take the shapes of a circle, an ellipse, and a dumbbell, respectively. Elongation and fusion of fibers are stopped at 10 power transfer cycles to get the observable polarization difference Δt at the same strength of CL_{eff} . The factor C is the coupling coefficient, and L_{eff} is the effective coupler length. The output power changes as the couplers are cooled after fabrication. The changes are probably caused by stress birefringence. However, it is reported [7] that stress birefringence is negligibly small compared with form birefringence. It appears that the measured polarization difference Δt is produced mainly by form birefringence. The polarization difference Δt has the following proportional relation.

$$\Delta t = K(C_x - C_y) L_{\text{eff}} \quad (1)$$

where K is a proportional constant depending on the cross-sectional shape, and C_x and C_y are the coupling coefficients for the x and y polarized fundamental modes, respectively. Well fused couplers whose aspect ratio is less than about 1.4 have the positive Δt , and C_x is greater than C_y . However, Δt is negative for weakly fused couplers with smaller aspect ratio than about 1.4, and C_x is less

than C_y . A change in slope around $l/d = 1.4$ is possibly generated by a constricted part in the middle of the cross section. Fig. 2 proves an interesting feature that the coupling coefficient difference $C_x - C_y$ changes from positive to negative when the aspect ratio is about 1.4. The fused coupler with the aspect ratio of about 1.4 is isotropic, and its cross section is elliptical at this point. This experimental result differs from the theoretical calculations [8], [9]. The computations suggest that couplers with an elliptical and dumbbell cross sections are isotropic when the aspect ratios are about 1.93 [8] and 1.8 [9], respectively. This discrepancy between theoretical and experimental results may be caused by the change of cross section along fabricated couplers and different sizes and shapes of cross section between both results.

Fig. 3 shows the period of spectral power transfer against the aspect ratio. The coupled power P_2 changes sinusoidally with wavelength λ . The period T is very nearly the same for x and y polarized lights, and decreases with aspect ratio except for a few points. The exceptions probably arise from uniformity of fusion and cleaving couplers at some distance from the coupler waist. Since the power transfer cycles are 10 for all fabricated couplers, the strength of CL_{eff} is the same at 1.3- μm wavelength. The change of CL_{eff} against wavelength for well-fused couplers is smaller than that for weakly fused ones.

Fig. 4 shows the ratio of polarization difference Δt to spectral power transfer period T . Since Δt is $T/2$ at $(C_x - C_y) L_{\text{eff}} = \pi/2$, the proportional constant in (1) is $K = T/2\pi$ and we can get an equation as follows:

$$\frac{\Delta t}{T} = \frac{(C_x - C_y) L_{\text{eff}}}{\pi} \quad (2)$$

Fused couplers become polarizing beamsplitters when the polarization difference ratio $\Delta t/T$ is ± 0.5 . In order to make a polarizing beamsplitter with fewer power transfer cycles, the aspect ratio should be large, i.e., the degree of fusion should be weak.

Fig. 5 shows the elongation length L for couplers with 10 power transfer cycles at 1.3- μm wavelength. The strength of CL_{eff} is the same at 1.3 μm for all test couplers. The length L_{eff} becomes shorter for well-fused couplers with less aspect ratio because coupling coefficient C for well-fused couplers is larger than that for weakly fused couplers. Power transfer begins at less elongation length for well-fused couplers. Therefore the difference $L - L_{\text{eff}}$ for well fused couplers is smaller than that for weakly fused couplers, i.e., L_{eff} increases slowly with aspect ratio compared with L .

Fig. 6 shows the tapered fiber diameter d at the taper waist for couplers with 10 power transfer cycles. Since couplers with large aspect ratio are weakly fused and the coupling is weak, the couplers are extended long to get the same power transfer cycles, and the constituent fibers come to be thin. The fiber diameter for the coupler with the aspect ratio of about 1.48 is relatively thick because the coupler is cut away from the coupler's waist. The aspect ratio, therefore, is presumably smaller than 1.48.

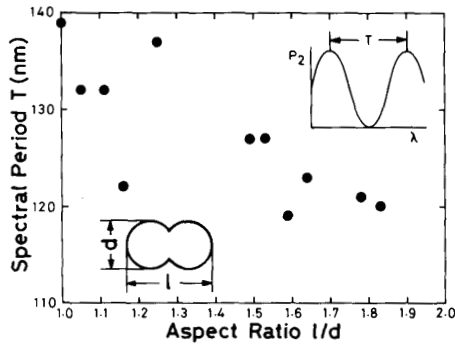


Fig. 3. Period of spectral power transfer against aspect ratio. The coupled power P_2 with wavelength λ and the period T are shown schematically.

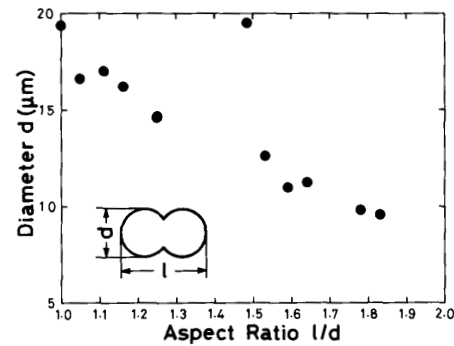


Fig. 6. Fiber diameter d at the coupler waist for couplers with 10 power transfer cycles at 1.3- μ m wavelength.

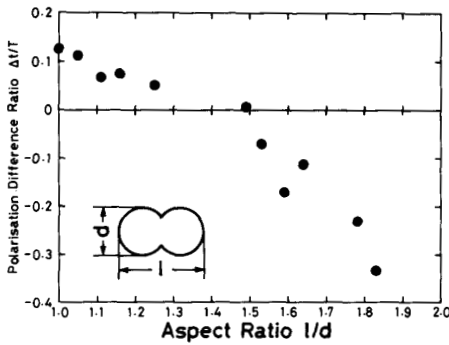


Fig. 4. Ratio of polarization difference Δt to spectral power transfer period T .

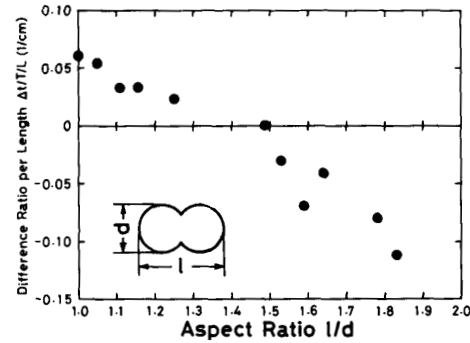


Fig. 7. Polarization difference ratio $\Delta t/T$ per elongation length L .

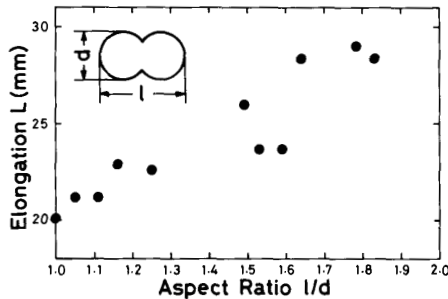


Fig. 5. Elongation length L for fused couplers with 10 power transfer cycles at 1.3- μ m wavelength.

Fig. 7 shows the polarization difference ratio $\Delta t/T$ per the elongation length L . It can be derived from (2) that the coupling coefficient difference between x and y polarized lights is in proportion to $\Delta t/T/L_{\text{eff}}$ as follows:

$$\frac{\Delta t}{TL_{\text{eff}}} \propto (C_x - C_y). \quad (3)$$

Since the difference $L - L_{\text{eff}}$ is great at large aspect ratio, $(\Delta t/T/L_{\text{eff}} - \Delta t/T/L)$ is enlarged more at large aspect ratio than at small one. The difference of coupling coef-

ficients between the x and the y polarized modes is comparatively large at aspect ratio of about 1.0 and more than 1.7. The sign of the difference at both aspect ratios is opposite to each other. The tendency of the coupling coefficient difference is similar to the theoretical analysis for fused couplers modeled by a dumbbell cross section [9]. However, the experiment presented here differs quantitatively from the analysis, and the polarization difference between coupling coefficients $C_x - C_y$ is very large above the aspect ratio of 1.7 contrary to the theoretical computations.

III. POLARIZING BEAMSPLITTERS

On the basis of experimental results shown in the previous section, polarizing beamsplitters are manufactured. We can fabricate polarizing beamsplitters with few power transfer cycles by adjusting the degree of fusion so that the aspect ratio is about 1.0 and more than 1.7 because the polarization difference ratio $\Delta t/T$ shown in Fig. 4 is relatively large at the aspect ratio of about 1.0 and more than 1.7. We stop elongation and fusion of couplers when fluctuation of output power monitored at coupling port becomes smallest. The scan length of the flame is 10 mm to achieve the thick coupler's waist.

Fig. 8 shows splitting ratios of a polarizing beamsplitter with the aspect ratio of 1.1 for x , y polarized, and

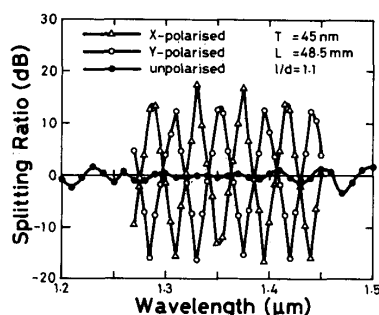


Fig. 8. Splitting ratios of a polarizing beamsplitter with the aspect ratio of 1.1 for x , y polarized and unpolarized lights.

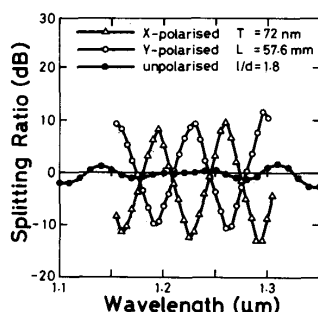


Fig. 9. Splitting ratios of a polarizing beamsplitter with the aspect ratio of 1.8 for x , y polarized and unpolarized lights.

unpolarized lights. The cross-sectional shape is almost circular. The loss is less than 0.6 dB in the region of 1.2- to 1.5- μm wavelength. The power transfer cycle is 28, the elongation length L is 48.5 mm, the period of spectral splitting ratio T is 45 nm, and the splitting ratios are 17.6 and -16.1 dB at 1.33 μm for x and y polarized lights, respectively. Considering monochromatic light with the theoretical spectral width of about 2.5 nm and the observation wavelength interval of 5 nm, it appears that polarization selective splitting of the beamsplitter is better than the measured values of 17.6 and -16.1 dB.

Fig. 9 shows splitting ratios of a polarizing beamsplitter whose aspect ratio is 1.8. The degree of fusion is considerably weak. The loss is less than 0.3 dB in the wavelength region of 1.1 to 1.35 μm . The elongation length L is 57.6 mm and the splitting ratios at 1.295 μm are -12.7 and 11.7 dB for x and y polarized lights. The power transfer cycle is 17 which is less than that of the former polarizing beamsplitter because the absolute polarizing difference ratio $|\Delta t/T|$ for the aspect ratio of 1.8 is greater than for the aspect ratio of 1.1 as shown in Fig. 4. The spectral period T is 72 nm and longer comparing with the well-fused polarizing beamsplitter shown in Fig. 8. Wavelengths where couplers serve as polarizing beamsplitters are discrete, and their spaces for weakly fused polarizing beamsplitters are greater than for well-fused ones.

IV. CONCLUSIONS

The polarization properties of fused tapered fiber couplers are examined experimentally. The wavelength difference of spectral splitting ratio between x and y polarized lights depends dramatically on the cross-sectional shape of fused couplers. The shape changes from a dumb-bell to an ellipse and a circle with increasing the degree of fusion. The difference of coupling coefficients between x and y polarized lights is relatively large for the aspect ratio of about 1.0 and more than 1.7. The coupling coefficient of the x polarized mode is greater and smaller than that of the y polarized mode for the aspect ratio of less and more than about 1.4, respectively. The couplers with the aspect ratio of about 1.4 is isotropic and insensitive to polarization state.

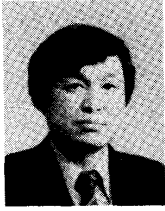
On the basis of the experimental results, two polarizing beamsplitters with the aspect ratio of 1.1 and 1.8 are produced, and their chromatic characteristics are measured. It turned evident that the aspect ratio should be more than 1.7 to manufacture polarizing beamsplitters with few power transfer cycles, i.e., long spectral period. Splitting ratios of polarizing beamsplitters with the aspect ratio of 1.1 and 1.8 are 17.6 and -12.7 dB for x polarized light and -16.1 and 11.7 dB for y polarized light, respectively. Their losses are less than 0.6 and 0.3 dB.

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